Protein Targeting and Degradation

Protein targeting and degradation

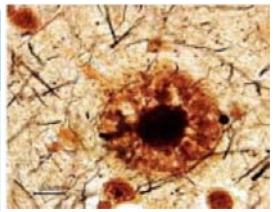
Learning goals:

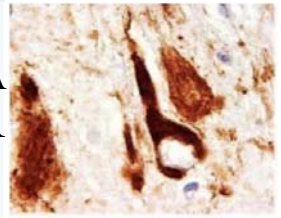
- Polypeptides fold by a stepwise process
- Some proteins undergo assited folding
- Defects in protein folding provide the molecular basis for a wide range of human disorders
- Posttranslational modifications of proteins begins in the ER
- Glycosilation plays a key role in protein targeting
  - Targeting to lysosomes
  - Targeting to nucleus
- Protein degradation mediated by ubiquitin and proteasome

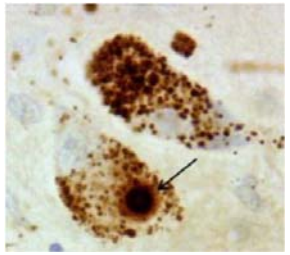
A common features in neurodegenerative diseases and others

Abnormal protein accumulation is a common feature in neurodegenerative diseases

Protein	Disease
β-amyloid	ALZHEIMER
Tau	ALZHEIMER PICK'S DISEASE FRONTOTEMPORAL DEMENTIA PROGRESSIVE SUPRANUCLEAR PALSY
α-synuclein	PARKINSON'S DISEASE
Huntingtin	HUNTINGTON'S DISEASE
CFTR	CYSTIC FIBROSIS



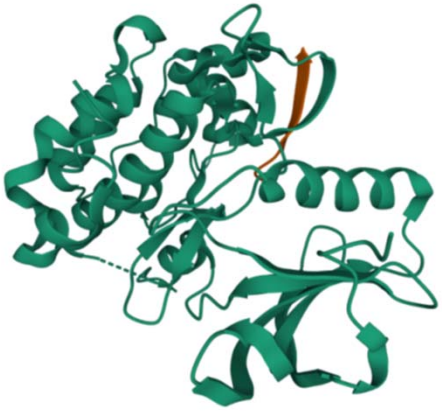






Proteins have specific tridimensional structures





Insulin-like Growth Factor

1Aa =10 different conformations 100 Aa =

10^{100} conformations

1 conformation = 10^{-13} s

Folding time for 100 Aa= 10^{87} s= 3.17 10^{79} year

The Dilemma: The age of the universe is estimated to be around 1.38 x10^{10} years. If protein folding were a purely random search, no protein in your body would ever fold. In reality, however, proteins fold into their active shapes within milliseconds (10^{-3} s).

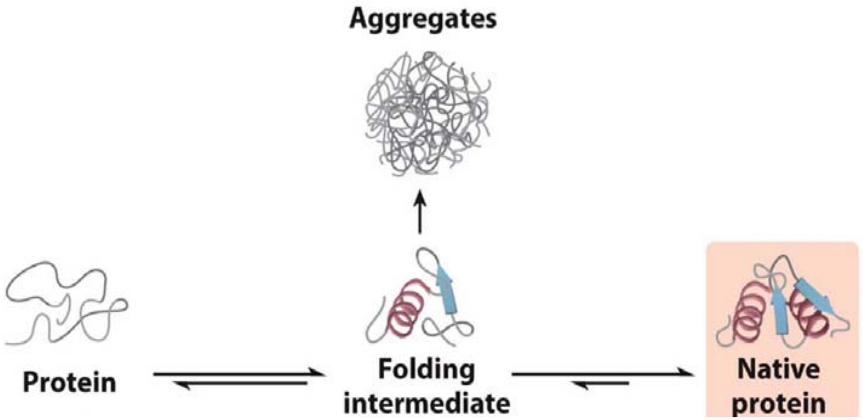
The Levinthal paradox

Protein folding

Polypeptides fold rapidly by a stepwise process



Chaperones Prevent Misfolding and Aggregation of Unfolded Peptides



Chaperonins Facilitate Folding

Chaperonin (GroEL/GroES)

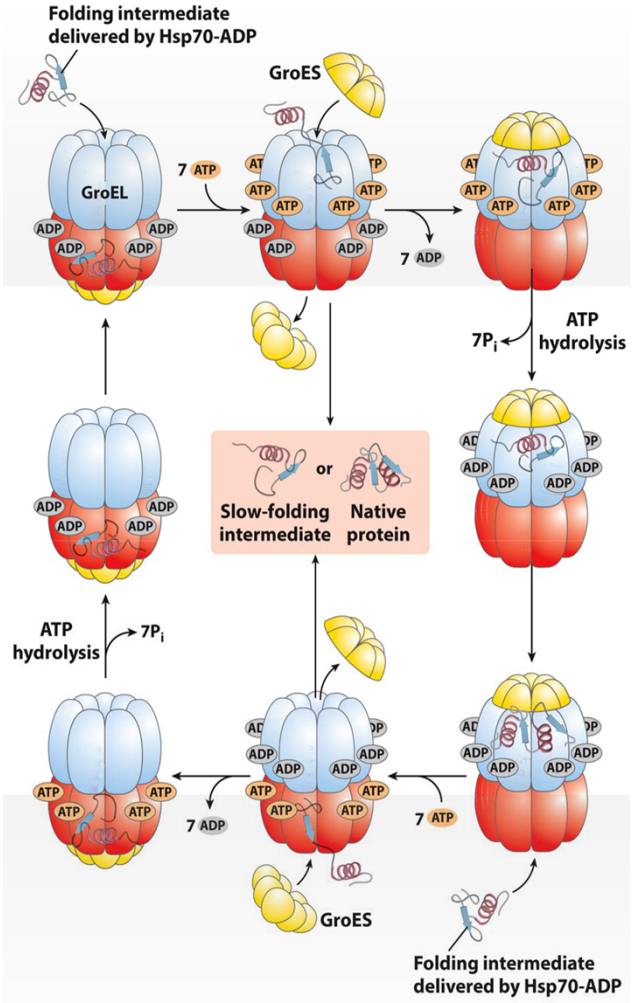
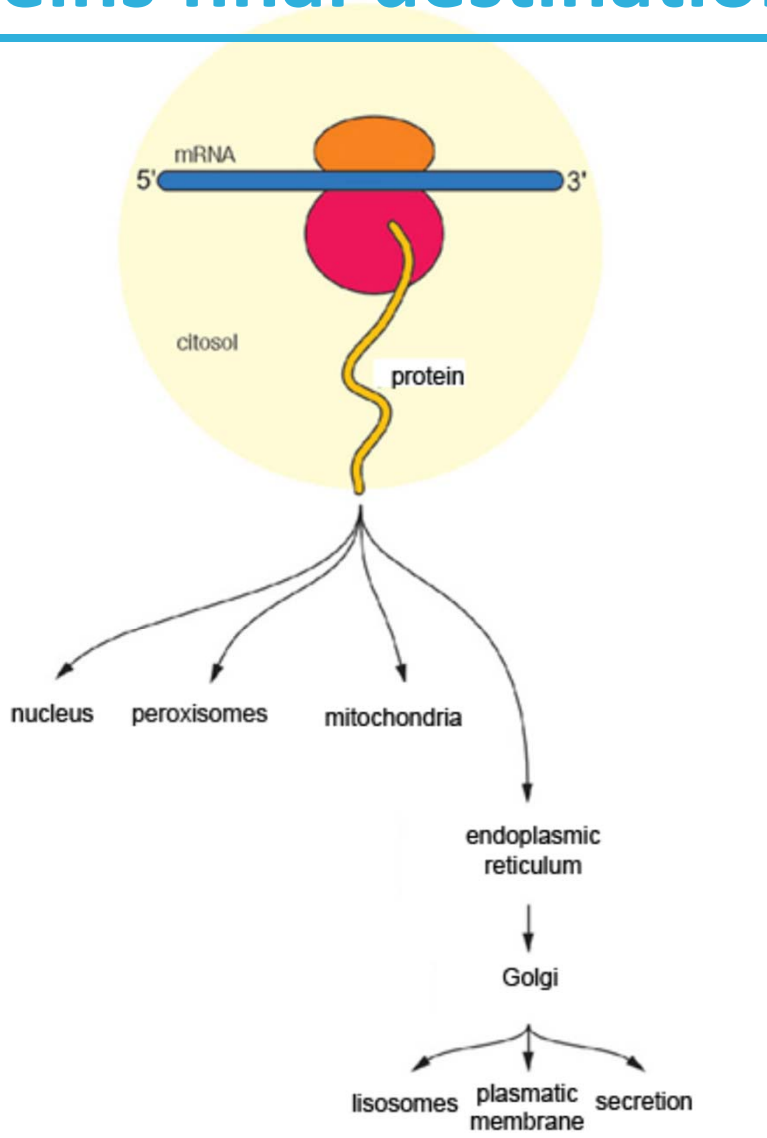


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Proteins final destinations



Protein classification depending on destination

Protein Targeting and Degradation

- Proteins move from site of synthesis to:
  - exit a cell
  - become part of the membrane
  - enter a subcellular compartment, and so on
- Most have a **signal sequence** at or near N-terminus
  - 13–36 amino acid residues in length
- This takes place in eukaryotic cells, where subcellular organization aids in compartmentalizing metabolic pathways.

Signal sequences that target proteins to the ER



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Signal characteristics:

- rich in hydrophobic Aa
- a positive charged Aa at the N-ter
- Aa with short side chain at C-ter (Ala, Gly)

Peptides Directed to the ER

- As peptide emerges from the ribosome, the signal sequence is bound by **signal recognition particle (SRP).**
- SRP/ribosome/RNA complex is delivered to the ER lumen.
  - Some modification takes place here (glycosylation, etc.).
- Transport vesicles then take proteins to Golgi apparatus.
- Proteins enter the biosynthetic/secretory pathway.
- Proteins for mitochondria bind chaperone proteins in the cytosol and are delivered to receptors on the exterior of the organelle.

Directing Peptides to the ER and Beyond

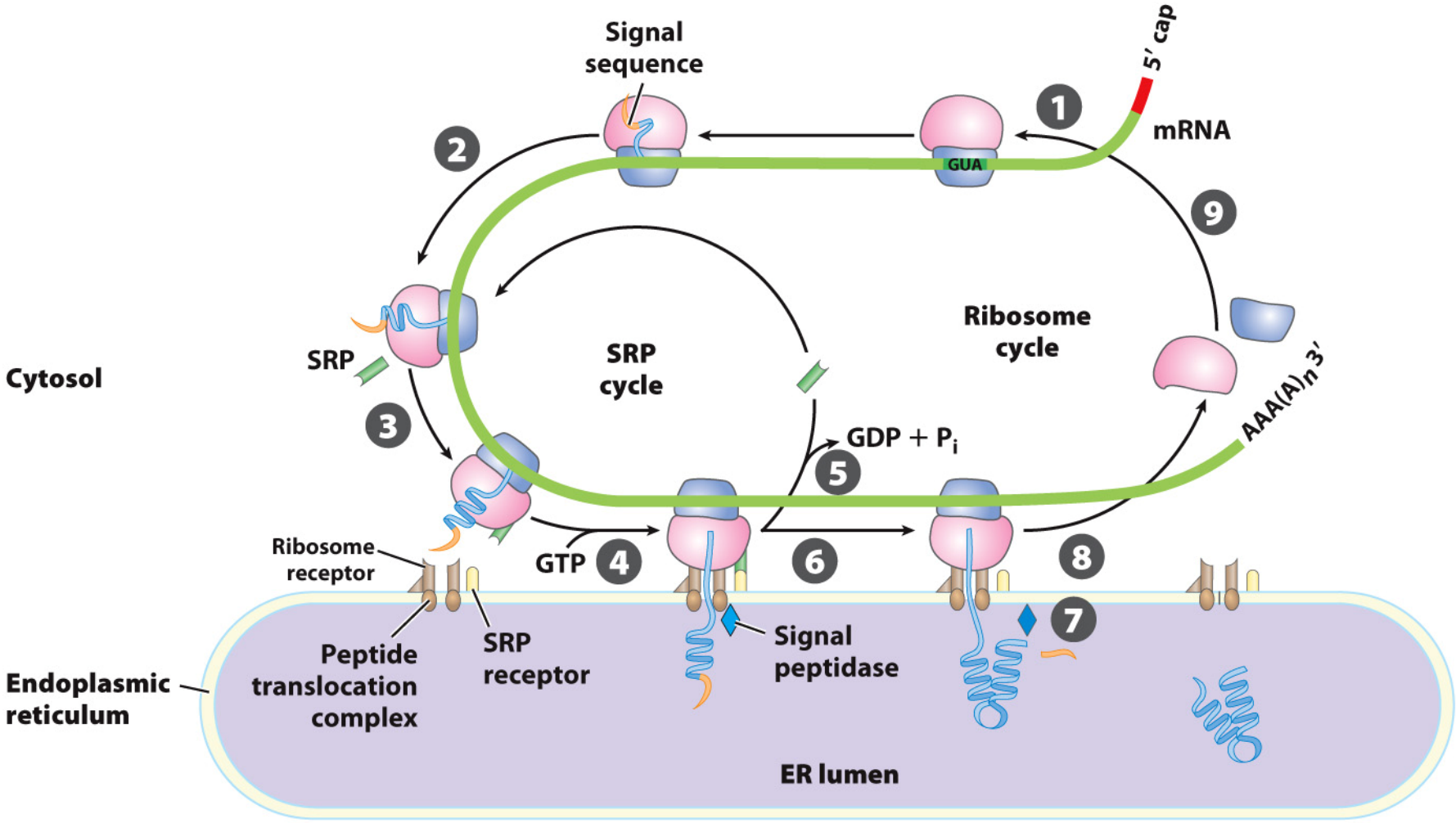
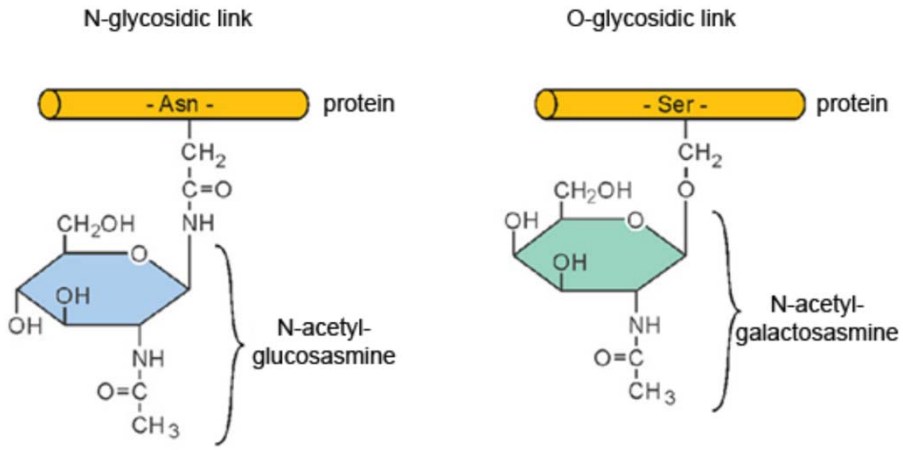


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Glycosylation of Proteins

- Glycoproteins form by linking an oligosaccharide to a side group on the peptide.



- N-linked (nitrogen of Asn or Arg)
  - begins in the RER
  - consesus sequence **NxS/T**

- O-linked (oxygen of Ser, Thr, Tyr)
  - typically occurs in the golgi or
    cytosol for proteins not entering the ER

  -consensus sequence **PxS/TxxP**

- Proteins are sorted in the Golgi and sent on to their final destinations.

Glycosylation of Proteins

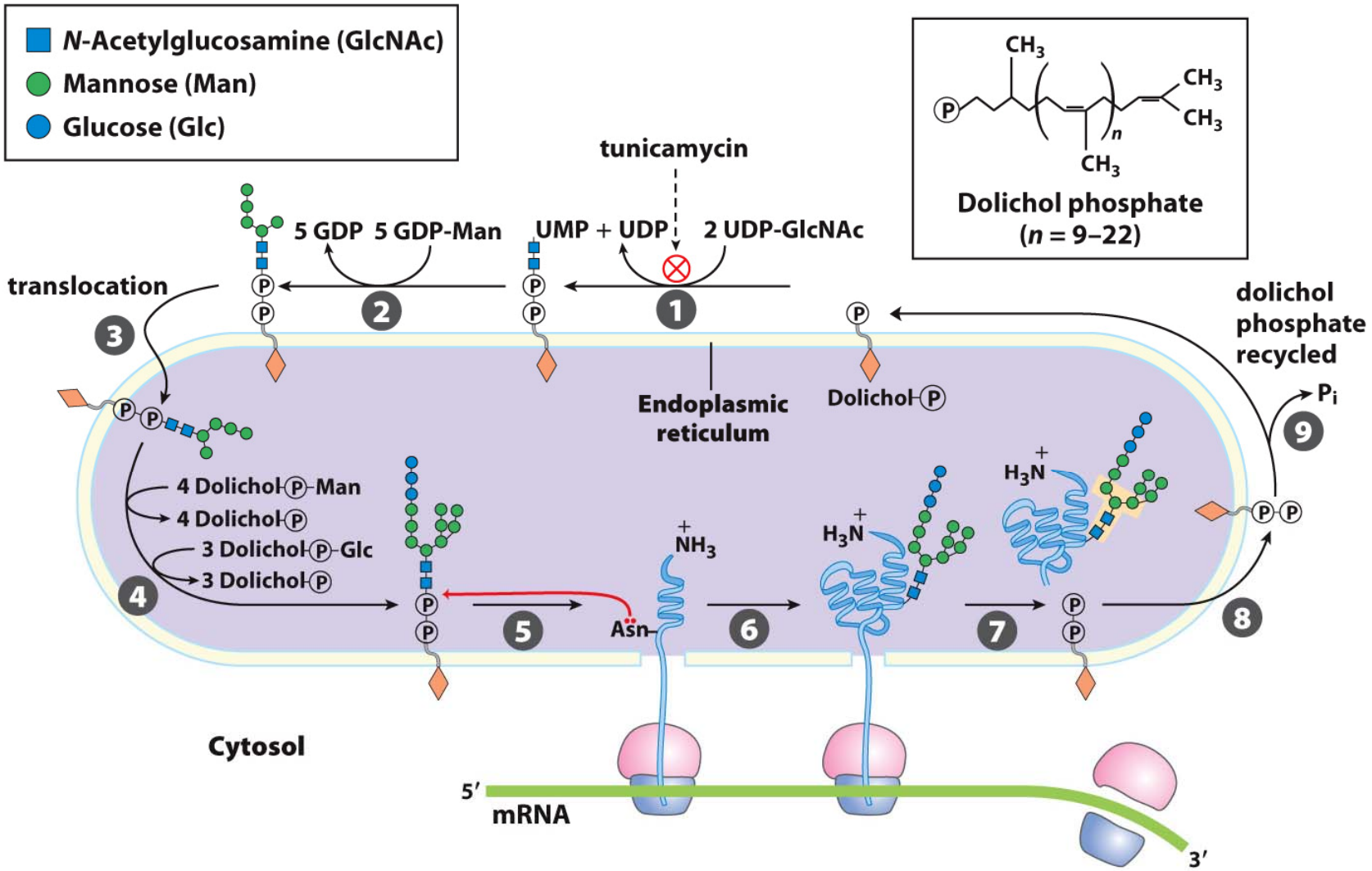


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Pathway taken by proteins destined for lysosomes

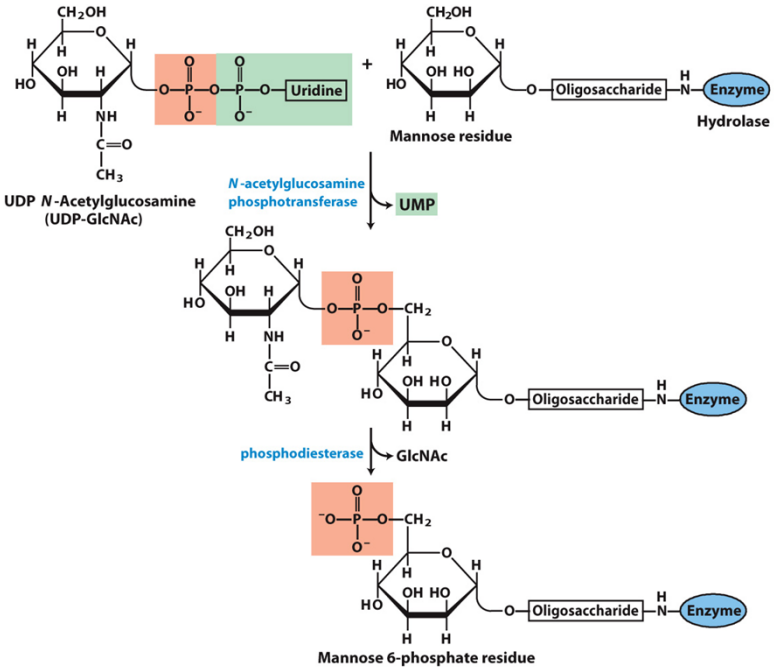
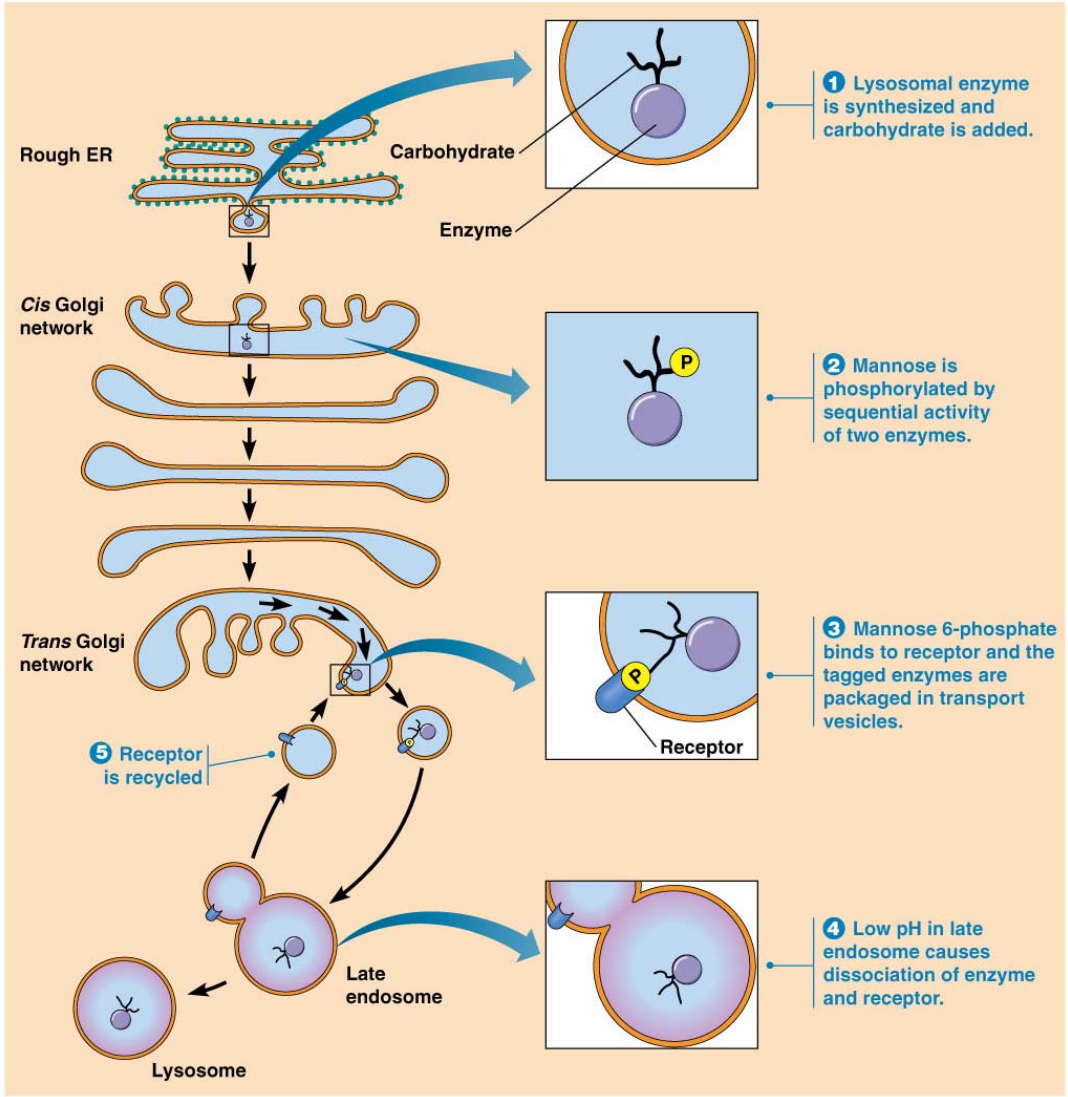


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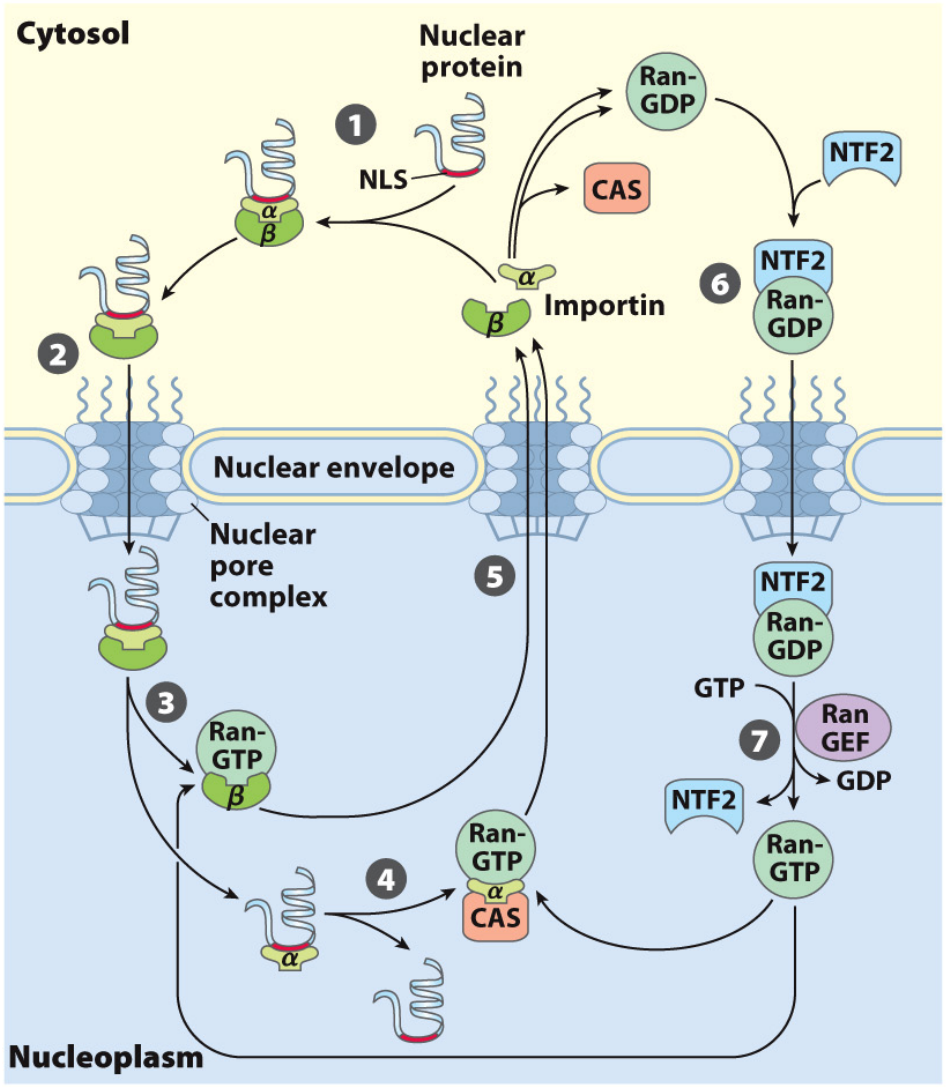
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How Proteins Are Targeted for and Imported into the Nucleus

- Proteins for the nucleus have a **nuclear localization sequence (NLS).**
- The NLS contains 4-8 basic aa (arginine or lysine).
- It binds importin α and β and a GTPase called Ran.
- The complex docks at a pore and is imported.
- Ribosomal proteins are synthesized in the cytosol, imported back into the nucleus, assembled into subunits in the nucleolus, and then exported back to the cytosol.

Targeting of Nuclear Proteins

(a)



(b)

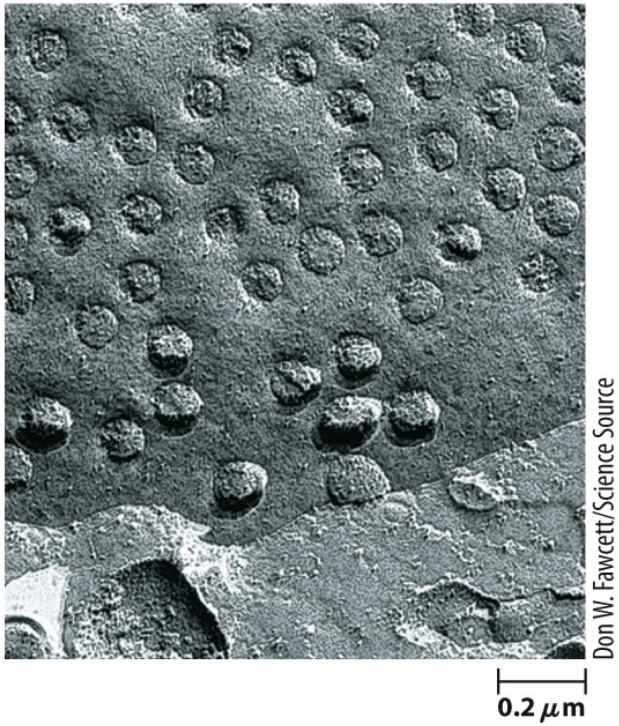


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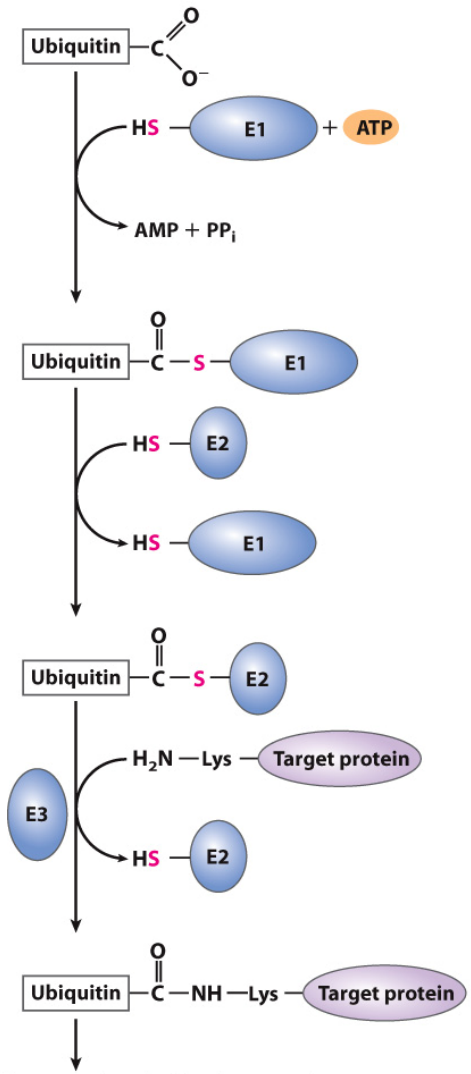
Protein Degradation Is Inevitable

- Half-life of proteins range from seconds to days to even months.
  - Hemoglobin is long lived.
  - Defective proteins are short lived, as are many metabolism regulatory proteins that respond to rapidly changing needs.
- But all are eventually degraded.

Mechanisms of Degradation

- In eukaryotes, proteins are linked to the protein **ubiquitin**.
  - via activating enzyme E1, conjugating enzyme E2, and ligating enzyme E3
- Ubiquinated proteins are cleaved by the 26s proteasome complex.
- Ubiquitin is very highly conserved among all eukaryotes.

Attachment of Ubiquitin to Target Proteins



Repeated cycles lead to attachment of additional ubiquitin.

Figure 27-49

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Relationchip between Protein Half-Life and Amino-terminal Amino Acid Residue

TABLE 27-9	Relationship between Protein Half-Life and Amino-Terminal Amino Acid Residue
Amino-terminal residue	**Half-life[a]**
Stabilizing	
Ala, Gly, Met, Ser, Thr, Val	> 20 h
Destabilizing	
Gln, Ile	~30 min
Glu, Tyr	~10 min
Pro	~7 min
Asp, Leu, Lys, Phe	~3 min
Arg	~2 min

Source: Information from A. Bachmair et al., *Science* 234:179, 1986.
[a]Half-lives were measured in yeast for the *β*-galactosidase protein modified so that in each experiment it had a different amino-terminal residue. Half-lives may vary for different proteins and in different organisms, but this general pattern appears to hold for all organisms.

PROTEASOME

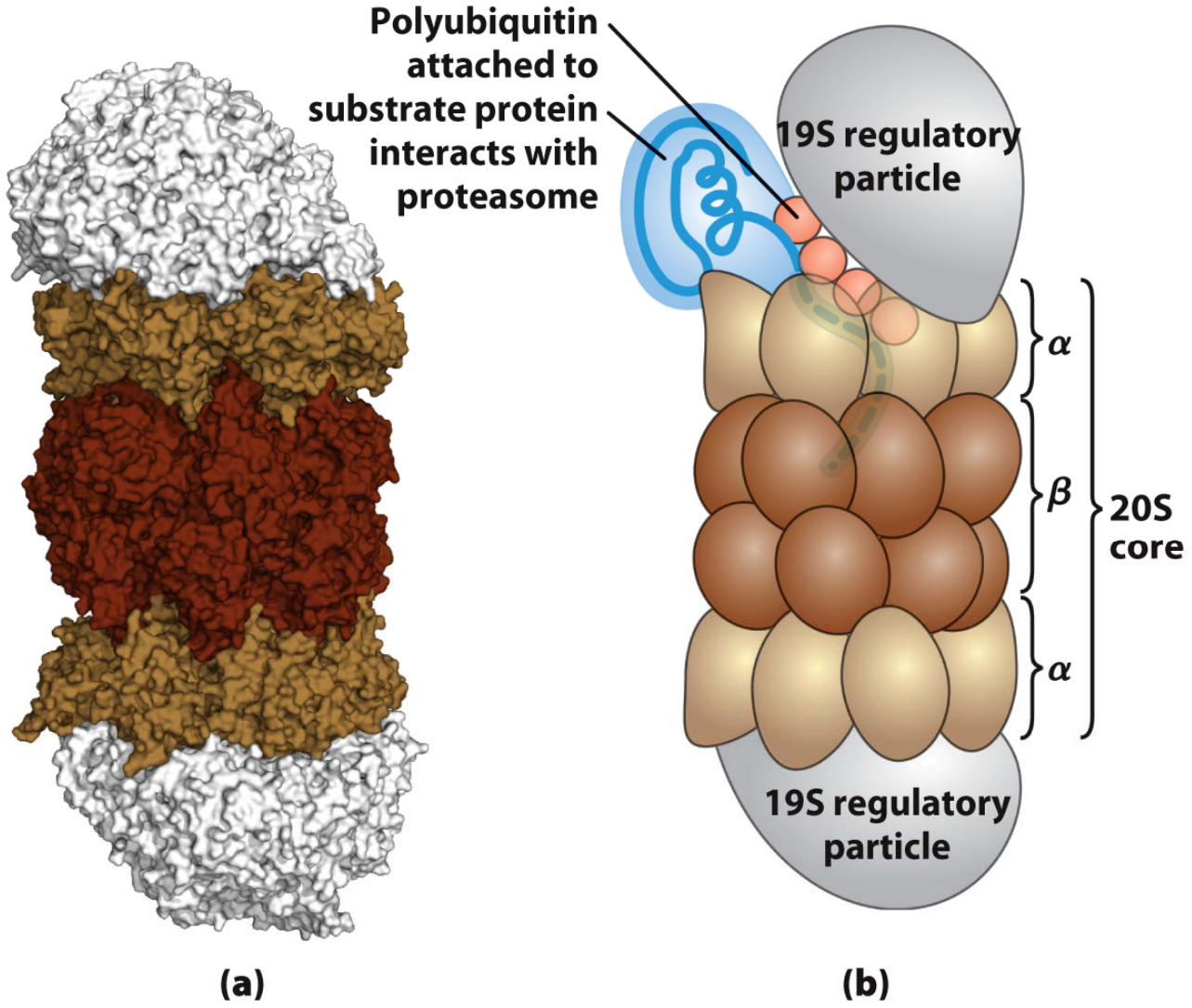


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The ubiquitin-proteasome system.

